

Two analytic statements that do not move RH: visibility at γ_1 , and why the 2-adic mass sits at $2^{\pm 1}$

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Abstract

Two calculations. (1) An off-line zero at height γ_1 makes the windowed form negative for every $\sigma \neq 0$ small, because the App. B term $\hat{f}(\rho)\hat{f}(1-\rho)$ opens at order $-\sigma^2$ on the ground state while $\lambda_{\min}$ is exponentially small. (2) The difference of Fourier transforms $F_S - F_\infty$ is a sum of dilations by $2^{\mathbb{Z}}$; conjugating the scaling $\vartheta(\lambda)$ against those dilations puts the extra trace mass at $\lambda = 2^{\pm 1}$, with leading weight the local Weil factor $(\log 2)/\sqrt{2}$. Neither statement is RH.

1 Visibility at γ_1

Let $\tau = L/2$, f even on $[-\tau, \tau]$, and

$$\hat{f}(\tfrac{1}{2} + \sigma + i\gamma) = \int_{-\tau}^{\tau} f(x) e^{\sigma x} e^{i\gamma x} dx.$$

App. B of *quorum-theorem*: $\hat{g}(\rho) = \hat{f}(\rho)\hat{f}(1-\rho)$ for $g = f \star \tilde{f}$. Expand at fixed γ in σ ,

$$\hat{f}(\tfrac{1}{2} + \sigma + i\gamma) = F_0 + \sigma F_1 + \tfrac{1}{2}\sigma^2 F_2 + O(\sigma^3),$$

with $F_0 = \hat{f}(\tfrac{1}{2} + i\gamma)$ real (even f) and $F_1 = \int f(x) x e^{i\gamma x} dx = iM$, $M = \int f(x) x \sin(\gamma x) dx$ real: for even f the first moment is purely imaginary. Then

$$\hat{g}(\rho) = F_0^2 - \sigma^2 |F_1|^2 + O(\sigma^2 F_0) + O(\sigma^4) = F_0^2 - \sigma^2 M^2 + \dots,$$

Take f the ground state of Q_L on V_N . Hyper-nullity (*zeros-from-the-radical*) gives $|F_0(\gamma_1)| \approx C_\gamma \lambda_{\min}$, exponentially small. The moment $|F_1(\gamma_1)|$ is $O(\tau \|f\|)$ a priori; it depends on N : on V_9 at $\mu = 11$ it is $O(1)$ ($M^2 \approx 0.54$ from $\hat{g} = -5.4 \times 10^{-5}$ at $\sigma = 0.01$), while on V_{47} at the same μ the same quantity, $|\partial_\gamma \hat{v}_0(\gamma_1)|$, is 3.5×10^{-3} and decays like $e^{-0.6\gamma}$ along the band (notebook §76): the visibility scale in σ moves by two orders of magnitude between the two bases. Hence on the V_9 vector

$$Q_L(f) + \hat{g}(\rho) = \lambda_{\min} - \sigma^2 |F_1(\gamma_1)|^2 + O(\sigma^2 \lambda_{\min}) + O(\sigma^4).$$

Lemma 1 (visibility at γ_1 , small σ). *If $F_1(\gamma_1) \neq 0$, then $Q_L + \hat{g}(\rho)$ is negative for every sufficiently small $\sigma \neq 0$.*

The hypothesis $F_1(\gamma_1) \neq 0$ is a non-vanishing of one Fourier moment of an explicit eigenvector. It is measured, not proved for every N, μ . It is the only analytic gap in the lemma.

High zeros. $|\hat{\eta}_0(\tfrac{1}{2} + \sigma + i\gamma)| \leq e^{|\sigma|\tau}/(|\gamma|\sqrt{L})$ and similarly for $n \geq 1$ away from ω_n . The extra Gram at height γ has operator norm $O(e^{2|\sigma|\tau}/\gamma^2)$ on V_N . At $\gamma = 80$, $\mu = 11$, this is 10^{-3} or less and does not move $\lambda_{\min}$ (measured: stays 10^{-22}). Visibility is local in height: a window of type τ sees an off-line zero near γ_1 , not one at height $\gg 2\pi N/L$. A single certified $Q_{11} \succ 0$ is compatible with an off-line zero at 10^6 . That is Weil on a compact window, not RH.

2 Why the mass sits at $2^{\pm 1}$

On the slice $\text{ord}_2 = 0$,

$$F_S = \frac{1}{2} \left(\sum_{n \geq 0} M_{2^n} - M_{1/2} \right) F_\infty,$$

where $(M_a \hat{g})(\rho) = \hat{g}(a\rho)$. Thus

$$F_S - F_\infty = \frac{1}{2} \sum_{n \geq 1} M_{2^n} F_\infty - \frac{1}{2} M_{1/2} F_\infty - \frac{1}{2} F_\infty.$$

The scaling $\vartheta(\lambda)g(r) = \lambda^{-1/2}g(r/\lambda)$ conjugates dilations of the frequency: $M_a \vartheta(\lambda) M_a^{-1} = \vartheta(\lambda/a)$ up to the normalization of $d^*\lambda$. Every extra copy M_{2^n} in the trace $\text{Tr}(\hat{P}_\Lambda P_\Lambda \vartheta(\lambda))$ produces a translate of the archimedean peak by $\lambda \mapsto 2^{\pm n}\lambda$. The first shells $n = \pm 1$ sit at $\lambda = 2$ and $\lambda = \frac{1}{2}$. Higher shells $2^{\pm n}$ are present in the formula and not resolved as isolated peaks at $\Lambda \leq 8$.

The coefficient of the first shell is the $k = 1$ term of the local Weil factor at $p = 2$: $(\log 2) 2^{-1/2} = (\log 2)/\sqrt{2}$. At $\Lambda = 4$ and $\Lambda = 8$ the hard-window mass of $\tau_S - \tau_\infty$ around $\lambda = 2$ extrapolates to that number as the cell size $h \rightarrow 0$ (0.488 and 0.491). That is a numerical limit of an explicit distribution, not a new identity. The archimedean peak at $\lambda = 1$ and the factor $2h(1) \log \Lambda$ are the other two terms of Connes (1999), Theorem 4; they are not re-derived here.

Remark. The hard window around $\lambda = \frac{1}{2}$ at $\Lambda = 8$ overshoots ($w = 1.17$ at 96 cells per unit): it swallows mass that is not the first shell. The peak at $\lambda = 2$ is clean. An analytic shell expansion would separate $n = \pm 1$ from $n = \pm 2, \pm 3$ without a window. That calculation is not done.

Status

Lemma 1 is a Taylor argument plus one measured non-vanishing. Proposition on the locus $2^{\pm 1}$ is conjugation of dilations; the weight is identified with the local factor and confirmed by extrapolation, not proved as an operator coefficient. No movement of RH. The two remaining analytic targets of the repository — $\delta_p \leq e^{-7p}$ and $\kappa_p \geq e^{-4p}$ on the bottom vector, uniformly in μ — are not touched.